MTH357

Specimen Examination Paper

Applied Regression Analysis I

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SIX (6)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **open-book** examination.

At the end of the examination
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

Answer all questions. (Total 100 marks)

Question 1

A researcher uses data for a sample of 88 houses sold in an urban area during a recent year to investigate the relationship between house prices y_j (in thousands of dollars) and house size x_j (in square metres) using this equation:

$$y_j = \hat{\beta}_0 + \hat{\beta}_1 x_j + \hat{\varepsilon}_j, \quad j = 1, 2, \ldots, N$$

where $\hat{\beta}_0$ and $\hat{\beta}_1$ *are the LS estimators of the intercept coefficient* β_0 *and the slope coefficient* β_1, $\hat{\varepsilon}_j$ *is the residual for the* j^{th} *sample observation, and N is sample size.*

Preliminary analysis of the sample data produces the following information:

$$\sum_{j=1}^{N} y_j = 25832.05, \quad \sum_{j=1}^{N} x_j = 16462.34, \quad \sum_{j=1}^{N} y_j^2 = 8500750.60, \quad \sum_{j=1}^{N} x_j^2 = 3329789.60,$$

$$\sum_{j=1}^{N} (y_j - \bar{y})^2 = 917854.51, \quad \sum_{j=1}^{N} (x_j - \bar{x})^2 = 250144.32, \quad \sum_{j=1}^{N} \hat{\varepsilon}_j^2 = 348053.43,$$

$$\sum_{j=1}^{N} x_j y_j = 5209990.70, \quad \sum_{j=1}^{N} (x_j - \bar{x})(y_j - \bar{y}) = 377534.76, \quad N = 88$$

Use the above information to answer all the following questions. Show explicitly all formulas and calculations.

(a) Determine the estimates of the intercept and slope parameters β_0 and β_1.

(7 marks)

(b) Interpret the slope parameter estimate $\hat{\beta}_1$ obtained in part (a), i.e. explain what the numeric value of $\hat{\beta}_1$ means.

(4 marks)

(c) Determine an estimate of the error variance σ^2.

(3 marks)

(d) Determine the value of the coefficient of determination, r^2, for the estimated regression equation. Briefly explain what the calculated value of r^2 means.

(6 marks)

Question 2

In some situations, a regression model with an intercept of zero may be fitted:

$$y_j = \beta_1 x_j + \varepsilon_j, \quad j = 1, 2, \ldots, n$$

As usual, the error terms are assumed to be independent of each other and have mean zero and constant variance σ^2 so that $E(\varepsilon_i \cdot \varepsilon_j) = 0$ for $i \neq j$ and $E(\varepsilon_j^2) = \sigma^2$.

(a) Determine the least-squares estimator for the slope parameter $\hat{\beta}_1$.

(7 marks)

(b) State what are the values of $\sum_{j=1}^{N} \hat{\varepsilon}_j$ and $\sum_{j=1}^{N} x_j \hat{\varepsilon}_j$. Briefly explain what they mean.

(5 marks)

(c) Consider the following two simple regression models.

Model I: $y_j = \beta_0 + \beta_1 x_j + \varepsilon_j$

Model II: $y_j = \alpha_0 + \alpha_1 x'_j + \varepsilon_j$, where $x'_j = x_j - \bar{x}$

(i) State the estimators of $\hat{\beta}_0$ and $\hat{\alpha}_0$. Are they identical? Do they have the same variance?

(7 marks)

(ii) State the estimators of $\hat{\beta}_1$ and $\hat{\alpha}_1$. Are they identical? Do they have the same variance?

(5 marks)

(iii) What is the advantage, if any, of Model II over Model I?

(1 mark)

(d) Suppose that you run the following regression

$$y'_j = \beta_0 + \beta_1 x'_j + \varepsilon_j$$

where $y'_j = y_j - \bar{y}$ and $x'_j = x'_j - \bar{y}$ are the deviations from their respective mean values. Find the value of $\hat{\beta}_0$. Hence interpret the regression line.

(5 marks)

Question 3

The following two regressions are conducted, where the covariates are Age (x1), Income (x2), Price(x3). Sales is represented by y.

$$\hat{y}_j = \underset{(1.53)}{28.60} + \underset{(10.82)}{0.258x_{2j}} - \underset{(1.32)}{11.39x_{3j}}$$

$$R^2 = 0.83$$

and

$$\hat{y}_j = \underset{(0.78)}{-11.74} + \underset{(5.38)}{0.986x_{1j}} + \underset{(2.30)}{0.084x_{2j}} - \underset{(0.08)}{0.531x_{3j}}$$

$$R^2 = 0.91$$

<u>Figures in brackets are the *t*-values for individual parameter estimates</u>.

(a) Given that the sample size is 55, perform global F-test on the 1st regression equation.

(8 marks)

(b) Do you have sufficient information to perform a nested model F-test to determine if the second model is superior to the first model? Justify. If it can be done, perform the test.

(6 marks)

(c) Discuss and interpret the regression equation and coefficients.

(6 marks)

Question 4

Creatinine clearance (Y) is a measure of kidney function but is difficult to obtain as it requires 24-hour urine collection. Thus, we would like to determine if we can estimate Y with data which is more easily available, i.e. X_1 – serum creatinine concentration, X_2 – age and X_3 - weight. The sample consists of 57 males.

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

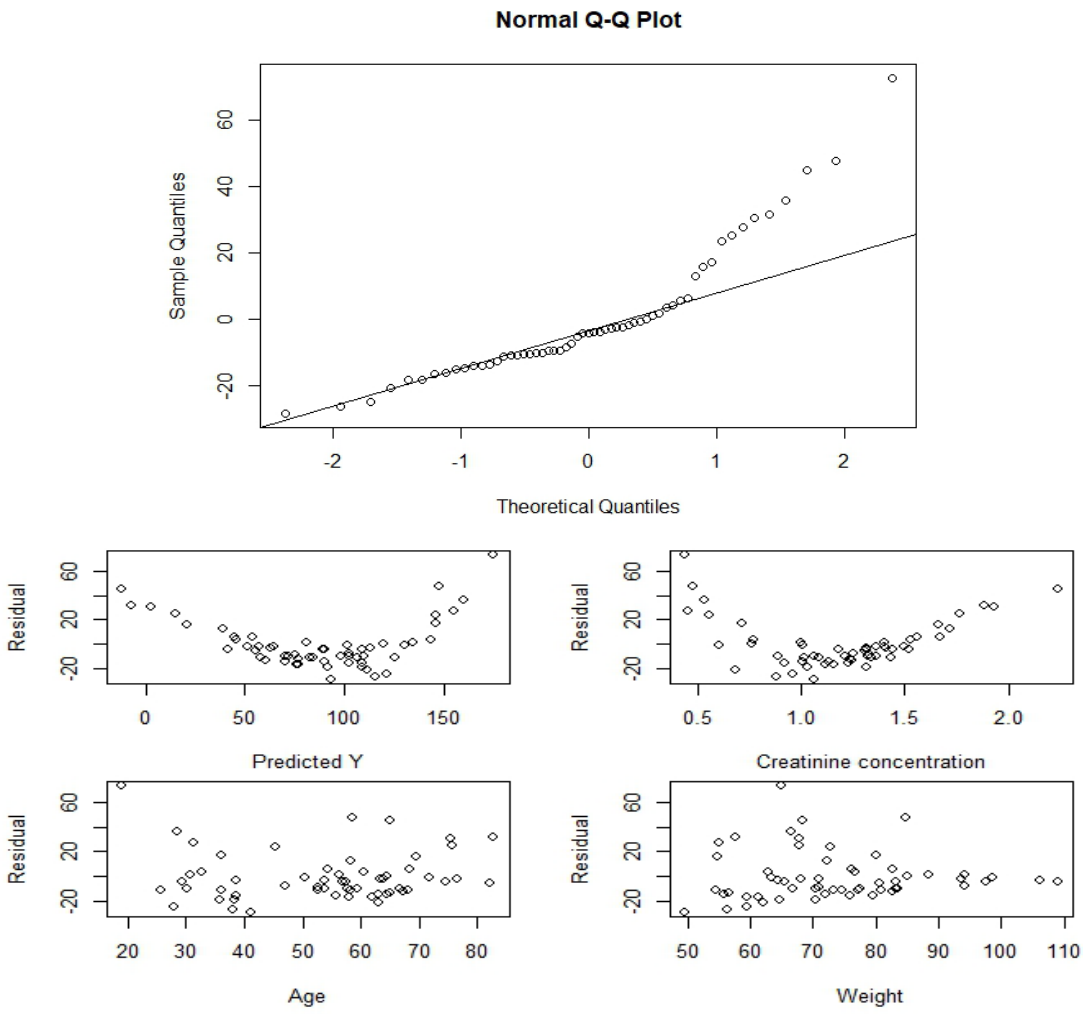

Figure Q4(a)

(a) Describe the 5 residual plots above and their implications.

(10 marks)

(b) Will you recommend to make a change to the regression model? Use transformations of data which can address the issues to make the model better.

(6 marks)

(c) Based on your recommendations in Q4(b), write down a plausible model to fit the data.

(3 marks)

Question 5

(a) The researcher has managed to get data on the school which each respondent is studying in. He wishes to apply a linear regression model that uses School, Age and Sleep per day as predictor variables for y, results for MTH357;

He has decided to set x_1 as:

$$x_1 = \begin{cases} 1, & \text{for School of Business} \\ 2, & \text{for SR Nathan School of Human Development} \\ 3, & \text{for School of Science and Technology} \\ 4, & \text{for other schools} \end{cases}$$

x_2: Age ($);
x_3: Sleep per day (Hours);

With the model:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

(i) Comment on the researcher's methodology for x_1 and interpret the meaning of β_1.

(5 marks)

(ii) Do you agree with this approach? Suggest how can the model be improved and interpreted.

(5 marks)

(b) The researcher decided to simplify the model by changing x_1, but instead include a new interaction term into the equation:

$$x_1 = \begin{cases} 1, & \text{or School of Science and Technology} \\ 0, & \text{if not} \end{cases}$$

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3$$

Interpret the model.

(6 marks)

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